Applied Numerical Methods for Civil Engineering

CGN 3405 - 0002

Week 7: Nonlinear Equations

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Quizzes Now!

- **Today's participation** (ungraded survey): Please check out

  "`Class Participation Quiz 13`"

  Time slot: **2:30PM – 3:00PM**

  on Canvas.

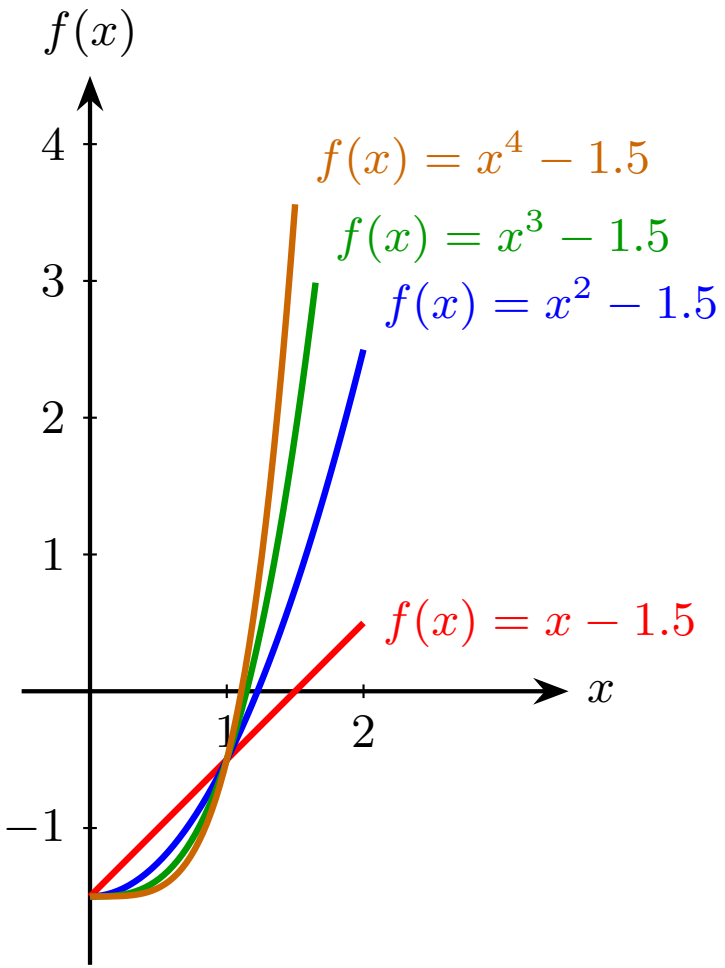
Learning Objectives

By the end of week, you will be able to:

- Derive the Newton-Raphson iteration formula from a Taylor series
- Apply the method to find roots of nonlinear equations
- Analyze the convergence behavior of the method
- Identify situations where the method may fail
- Implement the algorithm with Python programming

Problem Statement

- A nonlinear equation $f(x) = 0$
- Examples:
  - $x^2 - 2 = 0$ (Find $\pm\sqrt{2}$)
  - $\sin(x) = 0$ (Find multiples of π)
  - $e^{-x} - x = 0$ (Intersection of functions)
- Many engineering problems reduce to finding roots!



Historical Development

The modern method is a synthesis of three major mathematical contributions:

- **Isaac Newton (1669)**: Described method for polynomial equations
- **Joseph Raphson (1690)**: Formalized the iterative process
- **Thomas Simpson (1740)**: Generalized to calculus form using derivatives

Derivation from Taylor Series

- Expand $f(x)$ around x_n using Taylor series:

$$
\begin{aligned}
f(x_{n+1}) \approx & f(x_n) \\
& + f'(x_n)(x_{n+1} - x_n) \\
& + \frac{f''(x_n)}{2!}(x_{n+1} - x_n)^2 \\
& + \cdots \\
& + \frac{f^{(t)}(x_n)}{t!}(x_{n+1} - x_n)^t \\
= & \sum_{k=0}^{t} \frac{f^{(k)}(x_n)}{k!}(x_{n+1} - x_n)^k
\end{aligned}
$$

Derivation from Taylor Series

- Expand $f(x)$ around x_n using Taylor series:

$$
\begin{aligned}
f(x_{n+1}) \approx & f(x_n) \\
& + f'(x_n)(x_{n+1} - x_n) \\
& + \frac{f''(x_n)}{2!}(x_{n+1} - x_n)^2 \\
& + \cdots \\
& + \frac{f^{(t)}(x_n)}{t!}(x_{n+1} - x_n)^t \\
= & \sum_{k=0}^{t} \frac{f^{(k)}(x_n)}{k!}(x_{n+1} - x_n)^k
\end{aligned}
$$

- We want $f(x_{n+1}) = 0$
- Only use the first-order approximation:

$$f(x_{n+1}) = 0 \quad \Rightarrow \quad f(x_n) + f'(x_n)(x_{n+1} - x_n) \approx 0$$

Newton-Raphson Iteration Formula

- Let the first-order approximation be 0:

$$f(x_n) + f'(x_n)(x_{n+1} - x_n) = 0$$

- Rearrange the equation to solve for x_{n+1}:

$$\begin{aligned} & f'(x_n)(x_{n+1} - x_n) = -f(x_n) \\ \Rightarrow \quad & x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \\ \Rightarrow \quad & x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \end{aligned}$$

which is the Newton-Raphson iteration formula.

Newton-Raphson Iteration Formula

Understanding the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- x_n: Current estimate
- $f(x_n)$: Function value at current estimate
- $f'(x_n)$: Slope (first-order derivative) at current estimate
- $\frac{f(x_n)}{f'(x_n)}$: Correction term

Intuition: If $f(x_n)$ is large, we take a big step. If slope is steep, we take a smaller step.

Newton-Raphson Method

- Choose initial point x_0 and tolerance ε
- Set $n = 0$
- Repeat until convergence (using `while`)
  - Compute $f(x_n)$ and $f'(x_n)$
  - Calculate new estimate:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

  - Check convergence: If $\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < \varepsilon$, stop
  - Set $n = n + 1$

Stopping Criteria

Three common stopping criteria:

- Absolute error

$$|x_{n+1} - x_n| < \varepsilon$$

- **Relative error**

$$\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < \varepsilon$$

- Residual

$$|f(x_{n+1})| < \varepsilon$$

assume $f(x) = 0$ for some x

In practice, use a combination for robustness

Stopping Criteria

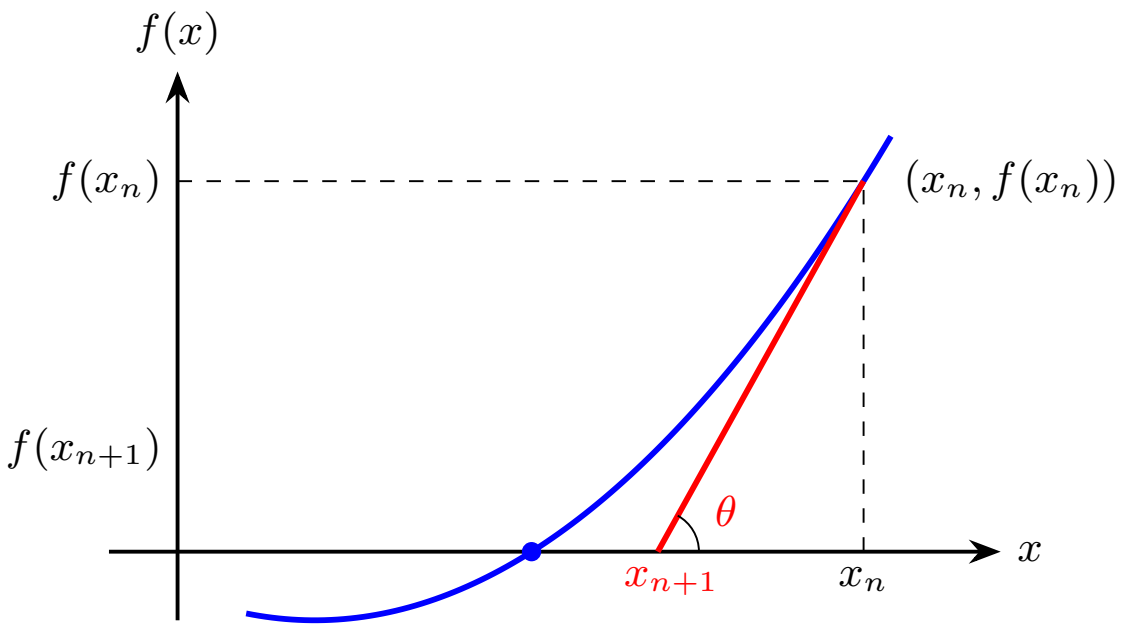
Always include a **maximum iteration limit** to prevent infinite loops

```
iter = 0
max_iter = 100
while error > tol and iter < max_iter:
    # Newton-Raphson iteration formula
    iter += 1
if iter >= max_iter:
    print('Warning: Maximum iterations reached')
```

Intuitive Understanding of Newton-Raphson Method

Given the **starting point** x_n for finding the root of the equation $f(x) = 0$

- Locate the point on the curve at this guess: $(x_n, f(x_n))$
- Draw the **tangent line** to the curve at this point
- Observe where this **tangent line crosses the x-axis**: $(x_{n+1}, 0)$

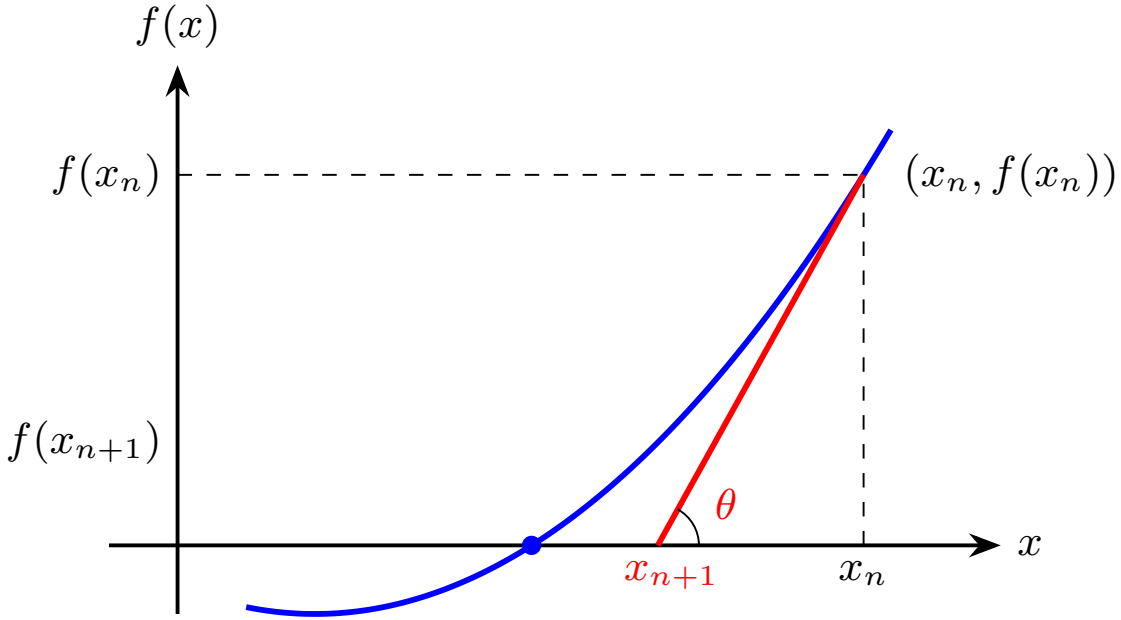


Intuitive Understanding of Newton-Raphson Method

Slope $f'(x_n)$:

- Derivative of function $f(x)$ evaluated at x_n
- Definition of the slope of a line:

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \frac{0 - f(x_n)}{x_{n+1} - x_n}$$



Intuitive Understanding of Newton-Raphson Method

We start with the equation:

$$f'(x_n) = \frac{-f(x_n)}{x_{n+1} - x_n}$$

1. Multiply both sides by $(x_{n+1} - x_n)$:

$$f'(x_n) \cdot (x_{n+1} - x_n) = -f(x_n)$$

2. Divide both sides by $f'(x_n)$:

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)}$$

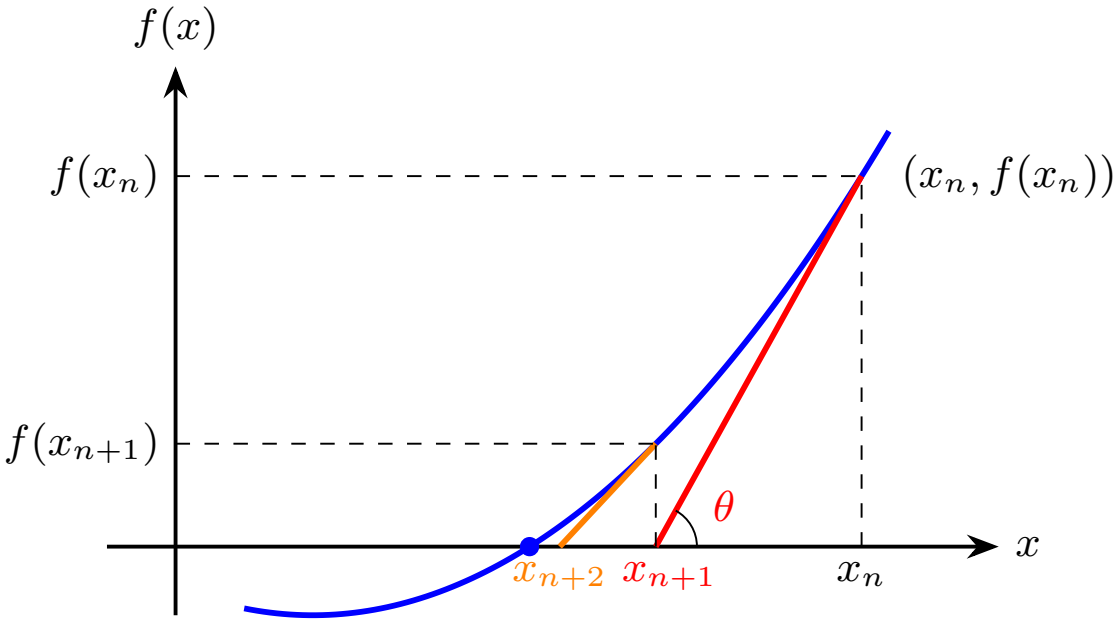
3. Add x_n to both sides to isolate x_{n+1}:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Intuitive Understanding of Newton-Raphson Method

How about slope $f'(x_{n+1})$?

$$f'(x_{n+1}) = \frac{0 - f(x_{n+1})}{x_{n+2} - x_{n+1}}$$



Intuitive Understanding of Newton-Raphson Method

How about slope $f'(x_{n+1})$?

$$f'(x_{n+1}) = \frac{0 - f(x_{n+1})}{x_{n+2} - x_{n+1}}$$

1. Multiply both sides by $(x_{n+2} - x_{n+1})$:

$$f'(x_{n+1}) \cdot (x_{n+2} - x_{n+1}) = -f(x_{n+1})$$

2. Divide both sides by $f'(x_{n+1})$:

$$x_{n+2} - x_{n+1} = -\frac{f(x_{n+1})}{f'(x_{n+1})}$$

3. Add x_{n+1} to both sides to isolate x_{n+2}:

$$x_{n+2} = x_{n+1} - \frac{f(x_{n+1})}{f'(x_{n+1})}$$

Solving $x^2 - 2 = 0$

Given initial point $x_0 = 1.5$, solve $x^2 - 2 = 0$

- Function $f(x) = x^2 - 2$
- Derivative $f'(x) = 2x$
- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n}$$

Solving $x^2 - 2 = 0$

Given initial point $x_0 = 1.5$, solve $x^2 - 2 = 0$

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- Newton-Raphson method:
  - First iteration: $x_1 = \frac{x_0}{2} + \frac{1}{x_0} = 1.416666667$
  - Second iteration: $x_2 = \frac{x_1}{2} + \frac{1}{x_1} = 1.414215686$
  - Third iteration: $x_3 = \frac{x_2}{2} + \frac{1}{x_2} = 1.414213562$
  - Fourth iteration: $x_4 = \frac{x_3}{2} + \frac{1}{x_3} = 1.414213562$

Converges to $\sqrt{2} = 1.414213562$ in just **3 iterations**!

Solving $x^2 - 2 = 0$

```python
def newton_raphson_sqrt2(x0, tol=1e-9, max_iter=100):
    n = 0
    xn = x0
    while n < max_iter:
        # Calculate f(xn) and f'(xn)
        fx = xn**2 - 2
        f_prime = 2 * xn
        # Newton-Raphson Formula
        x_next = xn - (fx / f_prime)
        # Calculate Relative Error
        error = abs(x_next - xn) / abs(x_next)
        # Check Convergence
        if error < tol:
            return x_next
        xn = x_next
        n += 1
    return xn

# Run with initial point x0 = 1.5
root = newton_raphson_sqrt2(1.5)
```

Quick Summary

Monday's Class:

- Newton-Raphson iteration formula: Derivation from Taylor series
- Intuitive understanding of Newton-Raphson method
- Solve the nonlinear equation $x^2 - 2 = 0$

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  on Canvas.

Solving $x^3 - 20 = 0$

How to solve $f(x) = x^3 - 20 = 0$ by Newton-Raphson method?

- First-order Taylor series approximation:

$$f(x_{n+1}) \approx f(x_n) + f'(x_n)(\underbrace{x_{n+1}}_{\text{unknown?}} - x_n)$$

- Let both sides be zero while x_{n+1} is unknown:

Solving $x^3 - 20 = 0$

How to solve $f(x) = x^3 - 20 = 0$ by Newton-Raphson method?

- First-order Taylor series approximation:

$$f(x_{n+1}) \approx f(x_n) + f'(x_n)(\underbrace{x_{n+1}}_{\text{unknown?}} - x_n)$$

- Let both sides be zero while x_{n+1} is unknown:

$$\begin{aligned} f(x_n) + f'(x_n)(x_{n+1} - x_n) &= 0 \\ f'(x_n)(x_{n+1} - x_n) &= -f(x_n) \\ x_{n+1} - x_n &= -\frac{f(x_n)}{f'(x_n)} \\ x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \end{aligned}$$

  which is the **Newton-Raphson iteration formula** at iteration n

- Initial point $x_0 = 3$

Solving $x^3 - 20 = 0$

How to solve $f(x) = x^3 - 20 = 0$ by Newton-Raphson method?

- Derivative $f'(x) = 3x^2$
- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 20}{3x_n^2}$$

- Initial point $x_0 = 3$
- **First iteration**:

$$\begin{aligned} x_1 &= x_0 - \frac{x_0^3 - 20}{3x_0^2} \\ &= 3 - \frac{3^3 - 20}{3 \times 3^2} \\ &= 3 - \frac{27 - 20}{27} = 3 - \frac{7}{27} = 2.74074 \end{aligned}$$

Solving $x^3 - 20 = 0$

How to solve $f(x) = x^3 - 20 = 0$ by Newton-Raphson method?

- Derivative $f'(x) = 3x^2$
- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 20}{3x_n^2}$$

- Initial point $x_0 = 3$
- **First iteration**:

$$\begin{aligned} x_1 &= x_0 - \frac{x_0^3 - 20}{3x_0^2} \\ &= 3 - \frac{3^3 - 20}{3 \times 3^2} \\ &= 3 - \frac{27 - 20}{27} = 3 - \frac{7}{27} = 2.74074 \end{aligned}$$

- **Relative error**:

$$\varepsilon_1 = \frac{|x_1 - x_0|}{|x_1|} = \frac{|2.74074 - 3|}{|2.74074|} = 9.46\%$$

Solving $x^3 - 20 = 0$

How to solve $f(x) = x^3 - 20 = 0$ by Newton-Raphson method?

- Derivative $f'(x) = 3x^2$
- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 20}{3x_n^2}$$

- Starting point $x_1 = 2.74074$
- **Second iteration**:

$$\begin{aligned} x_2 &= x_1 - \frac{x_1^3 - 20}{3x_1^2} \\ &= 2.74074 - \frac{2.74074^3 - 20}{3 \times 2.74074^2} \\ &= 2.74074 - 0.02607 = 2.71467 \end{aligned}$$

- **Relative error**:

$$\varepsilon_2 = \frac{|x_2 - x_1|}{|x_2|} = \frac{|2.71467 - 2.74074|}{|2.71467|} = 0.96\%$$

Solving $x^3 - 20 = 0$

How to solve $f(x) = x^3 - 20 = 0$ by Newton-Raphson method?

- Derivative $f'(x) = 3x^2$
- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 20}{3x_n^2}$$

- Starting point $x_2 = 2.71467$
- **Second iteration**:

$$\begin{aligned} x_3 &= x_2 - \frac{x_2^3 - 20}{3x_2^2} \\ &= 2.71467 - \frac{2.71467^3 - 20}{3 \times 2.71467^2} \\ &= 2.71467 - 0.00025 = 2.71442 \end{aligned}$$

- **Relative error**:

$$\varepsilon_3 = \frac{|x_3 - x_2|}{|x_3|} = \frac{|2.71442 - 2.71467|}{|2.71442|} = 0.009\%$$

Solving $x^3 - 20 = 0$

```python
def newton_raphson_x_cubic(x0, tol=1e-9, max_iter=100):
    n = 0
    xn = x0
    while n < max_iter:
        # Calculate f(xn) and f'(xn)
        fx = xn**3 - 20
        f_prime = 3 * xn**2
        # Newton-Raphson Formula
        x_next = xn - (fx / f_prime)
        # Calculate Relative Error
        error = abs(x_next - xn) / abs(x_next)
        print(x_next)
        print(error * 100)
        print()
        # Check Convergence
        if error < tol:
            return x_next
        xn = x_next
        n += 1
    return xn

x = newton_raphson_x_cubic(3)
```

Quadratic Convergence

- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- Let the root of $f(x) = 0$ be $x^\star$ (actual root)
- Add $-x^\star$ to both sides:

$$\underbrace{x_{n+1} - x^\star}_{\text{residual } \varepsilon_{n+1}} = \underbrace{x_n - x^\star}_{\text{residual } \varepsilon_n} - \frac{f(x_n)}{f'(x_n)}$$

or

$$\varepsilon_{n+1} = \varepsilon_n - \frac{f(x_n)}{f'(x_n)}$$

with residual

$$\varepsilon_n = x_n - x^\star$$

Quadratic Convergence (Optional)

- Expand $f(b)$ around a using a second-order Taylor polynomial:

$$f(b) \approx f(a) + f'(a)(b-a) + \frac{f''(\beta)}{2!}(b-a)^2$$

for some β between a and b.

- Expand $f(x^\star)$ around starting point x_n (defining residual $\varepsilon_n = x_n - x^\star$):

$$f(x^\star) = f(x_n) + f'(x_n)\underbrace{(x^\star - x_n)}_{=-\varepsilon_n} + \frac{f''(\beta)}{2!}\underbrace{(x^\star - x_n)^2}_{=\varepsilon_n^2}$$

where β is some value between $x^\star$ and x_n.

Quadratic Convergence (Optional)

- Expand $f(b)$ around a using a second-order Taylor polynomial:

$$f(b) \approx f(a) + f'(a)(b-a) + \frac{f''(\beta)}{2!}(b-a)^2$$

for some β between a and b.

- Expand $f(x^\star)$ around starting point x_n (defining residual $\varepsilon_n = x_n - x^\star$):

$$f(x^\star) = f(x_n) + f'(x_n)\underbrace{(x^\star - x_n)}_{=-\varepsilon_n} + \frac{f''(\beta)}{2!}\underbrace{(x^\star - x_n)^2}_{=\varepsilon_n^2}$$

where β is some value between $x^\star$ and x_n.

- Since $f(x^\star) = 0$ and $\varepsilon_n = x_n - x^\star$, we have

$$0 = f(x_n) - f'(x_n)\varepsilon_n + \frac{f''(\beta)}{2}\varepsilon_n^2$$

$$\Rightarrow \quad 0 = \frac{f(x_n)}{f'(x_n)} - \varepsilon_n + \frac{f''(\beta)}{2f'(x_n)}\varepsilon_n^2 \quad \Rightarrow \quad \varepsilon_n - \frac{f(x_n)}{f'(x_n)} = \frac{f''(\beta)}{2f'(x_n)}\varepsilon_n^2$$

Quadratic Convergence (Optional)

- Recall that

$$\varepsilon_n = x_n - x^\star$$

$$\varepsilon_{n+1} = \varepsilon_n - \frac{f(x_n)}{f'(x_n)}$$

we have

$$\varepsilon_n - \frac{f(x_n)}{f'(x_n)} = \frac{f''(\beta)}{2f'(x_n)}\varepsilon_n^2 \quad \Rightarrow \quad \varepsilon_{n+1} = \frac{f''(\beta)}{2f'(x_n)}\varepsilon_n^2$$

Quadratic Convergence

- Quadratic convergence established!

$$|\varepsilon_{n+1}| = \left| \frac{f''(\beta)}{2f'(x_n)} \right| \cdot \varepsilon_n^2$$

  As x_n approaches $x^\star$, the term $\left| \frac{f''(\beta)}{2f'(x_n)} \right|$ approaches a constant C

- Asymptotic error constant:

$$C = \left| \frac{f''(x^\star)}{2f'(x^\star)} \right|$$

  - C determines the speed of convergence
  - Small C implies faster convergence
  - If $f''(x^\star) = 0$, convergence can be even faster

- The order of convergence is 2:

$$|\varepsilon_{n+1}| \approx C \cdot \varepsilon_n^2$$

Quadratic Convergence

- Asymptotic error constant:

$$C = \left| \frac{f''(x^\star)}{f'(x^\star)} \right|$$

Or

$$|\varepsilon_{n+1}| \approx C \cdot \varepsilon_n^2 \quad \Rightarrow \quad \frac{|\varepsilon_{n+1}|}{|\varepsilon_n|^2} = \frac{|x_{n+1} - x^\star|}{|x_n - x^\star|^2} \approx C$$

- Example:

$$f(x) = x^2 - 2$$

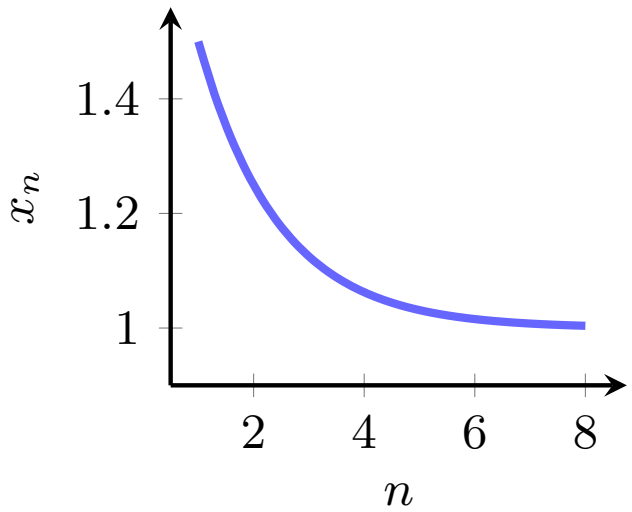
  - Find the root $f(x) = 0 \quad \Rightarrow \quad x^\star = \sqrt{2}$
  - Since $f'(x) = 2x$, we have $f'(x^\star) = 2\sqrt{2}$
  - Since $f''(x) = 2$, we have $f''(x^\star) = 2$
  - As a result, we have

$$C = \left| \frac{2}{4\sqrt{2}} \right| \approx 0.3535$$

Rate of Convergence

Linear convergence

$$\frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|} \leq r,\ r \in (0, 1)$$

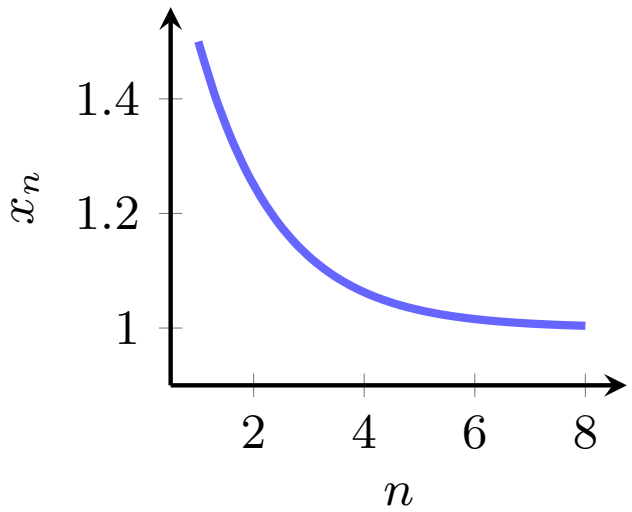


Sequence $x_n = 1 + \left(\frac{1}{2}\right)^n$ converges linearly to $x_\infty = 1$ because $r = \frac{1}{2}$

Rate of Convergence

Linear convergence

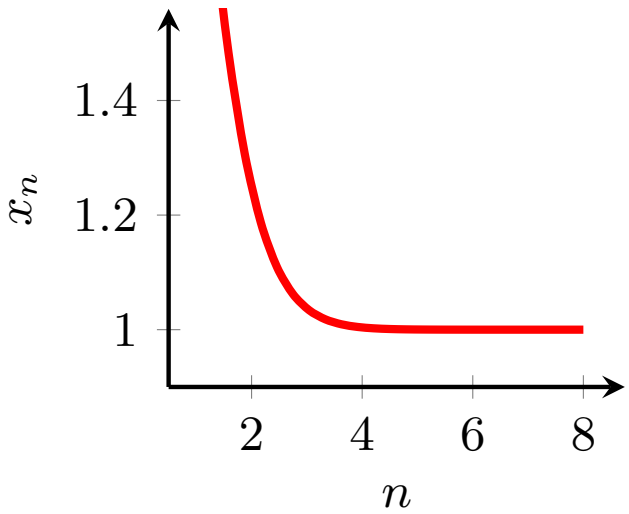
$$\frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|} \leq r,\ r \in (0,1)$$



Sequence $x_n = 1 + \left(\frac{1}{2}\right)^n$ converges linearly to $x_\infty = 1$ because $r = \frac{1}{2}$

Superlinear convergence

$$\lim_{n\to\infty} \frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|} = 0$$

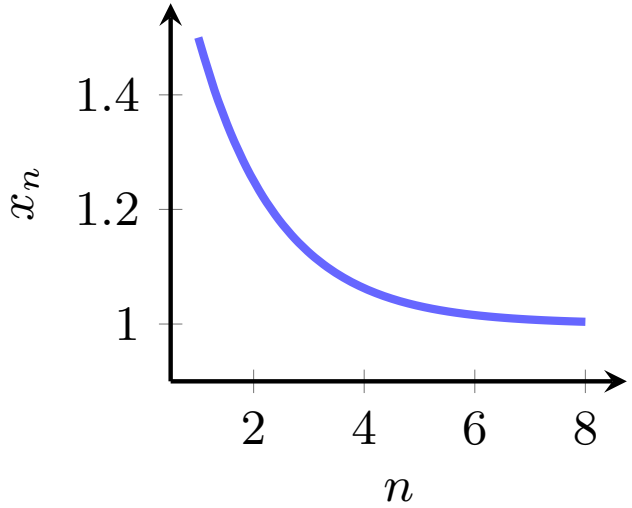


Sequence $x_n = 1 + \left(\frac{1}{n}\right)^n$ converges superlinearly to $x_\infty = 1$

Rate of Convergence

Linear convergence

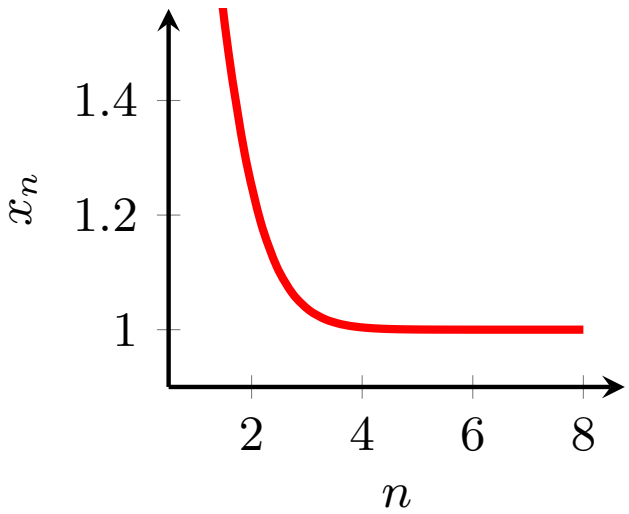
$$\frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|} \leq r,\ r \in (0,1)$$



Sequence $x_n = 1 + \left(\frac{1}{2}\right)^n$ converges linearly to $x_\infty = 1$ because $r = \frac{1}{2}$

Superlinear convergence

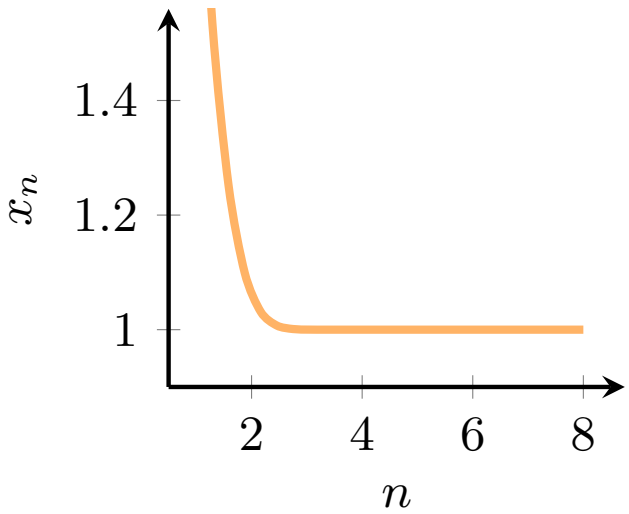
$$\lim_{n\to\infty} \frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|} = 0$$



Sequence $x_n = 1 + \left(\frac{1}{n}\right)^n$ converges superlinearly to $x_\infty = 1$

Quadratic convergence

$$\frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|^2} \leq M$$

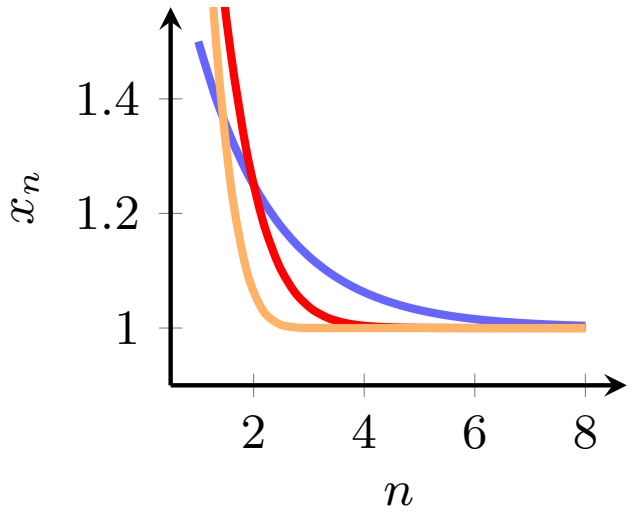


Sequence $x_n = 1 + \left(\frac{1}{n}\right)^{2^n}$ converges quadratically to $x_\infty = 1$ because $M = 1$

Rate of Convergence

Linear convergence

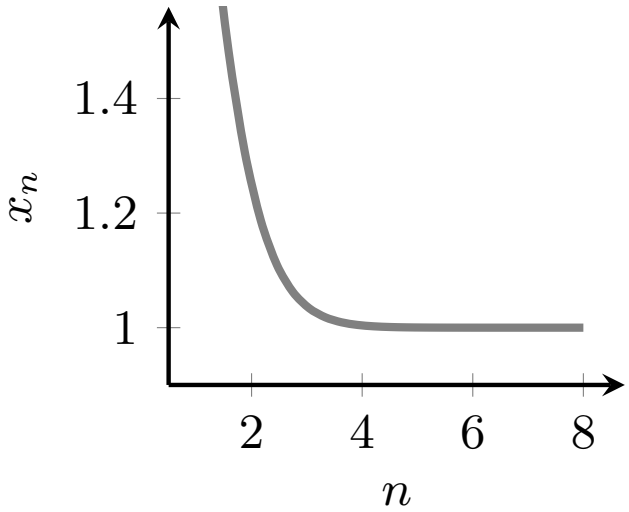
$$\frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|} \leq r,\ r \in (0,1)$$



Sequence $x_n = 1 + \left(\frac{1}{2}\right)^n$ converges linearly to $x_\infty = 1$ because $r = \frac{1}{2}$

Superlinear convergence

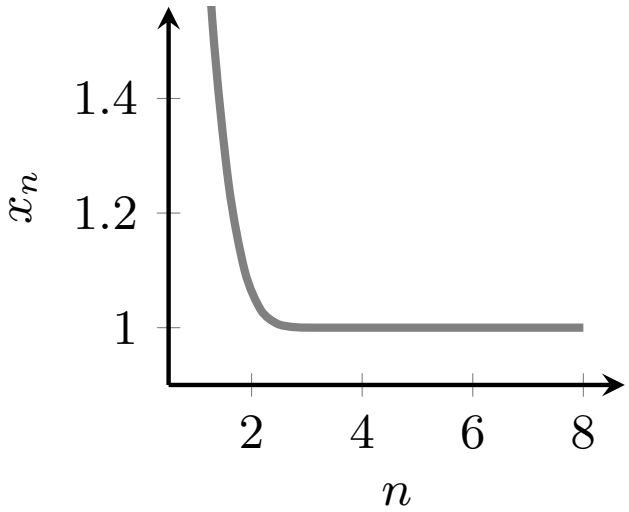
$$\lim_{n\to\infty} \frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|} = 0$$



Sequence $x_n = 1 + \left(\frac{1}{n}\right)^n$ converges superlinearly to $x_\infty = 1$

Quadratic convergence

$$\frac{|x_{n+1} - x_\infty|}{|x_n - x_\infty|^2} \leq M$$



Sequence $x_n = 1 + \left(\frac{1}{n}\right)^{2^n}$ converges quadratically to $x_\infty = 1$ because $M = 1$

Quick Summary

Wednesday's Class:

- Solve $f(x) = x^3 - 20 = 0$ by Newton-Raphson method
- Describe the quadratic convergence of Newton-Raphson method
- Compare linear, superlinear, and quadratic convergence